

Omar Nabail

omargamal039@gmail.com

| github.com/OmarNabail

| linkedin.com/in/omar-nabail

+49 152 25414448

PROFILE

M.Sc. Artificial Intelligence & Robotics candidate with hands-on experience building performance-focused image and data-processing systems in Python and C/C++. Applied LZ4 compression, distributed monitoring, OpenCV, signal processing, GPU-accelerated ML, API development, and automated test design.

EDUCATION

UTN – M.Sc. Artificial Intelligence & Robotics

Sep 2024 – Present

- **Master's Thesis: Text-Guided Scientific Figure Generation** – building and evaluating a routing system for executable Python, TikZ, and direct-image outputs using execution checks, visual metrics, VLM judges, and human evaluation.

German University in Cairo – B.Sc. Computer Science and Engineering

Sep 2017 – Sep 2023

- **Bachelor's Thesis: Valve Condition Monitoring** – developed an acoustic anomaly-detection pipeline using DFT, spectrogram analysis, MobileFaceNet, and sub-cluster AdaCos; achieved 81.38% AUC and 72.29% pAUC under domain shift.

EXPERIENCE

NLLG Lab, UTN – Student Research Assistant

Jul 2026 – Sep 2026

- Evaluate AI-generated images, captions, and code using structured quality criteria; identify, reproduce, and document failure cases across multimodal research projects.

UTN / Revel8 – Industry Collaboration: ML for Cybersecurity

Mar 2025 – Feb 2026

- Built a Python data-processing and classification pipeline combining parser-first preprocessing, local/cloud inference, and fine-tuned models; reduced input data volume by 90.2% and analyzed model failure cases.

SELECTED PROJECTS

Cluster Performance Analysis

Python, Kafka, DuckDB, LZ4

- Built decentralized monitoring pipelines for performance analytics across cluster hosts and clients; applied batching and LZ4 compression to improve data-transfer efficiency and reliability.
- Separated and encrypted data streams for different access patterns, supporting repeatable analysis of system behavior and reporting needs.

CAPTCHA Image Recognition – UTN Computer Vision Challenge

Python, PyTorch, OpenCV

- Designed, trained, and tested a CNN–Transformer model with deformable convolutions and triplet attention; analyzed recognition errors and placed 3rd of 8 teams.

PlotCraft: Tested Text-to-Python Visualization System

Python, FastAPI, Docker

- Built an API for executable Python generation and added sandboxed execution, pytest coverage, and GitHub Actions CI/CD for validation, Docker builds, and deployment ([GitHub](#)).

SKILLS

- **Programming:** C/C++, Python, SQL, Java, JavaScript
- **GPU & Distributed Workloads:** CUDA-enabled PyTorch, Slurm, distributed data pipelines
- **Image & Data Processing:** OpenCV, NumPy, Matplotlib, DFT, spectrograms, LZ4 compression, Kafka, DuckDB
- **Performance & Testing:** Performance analysis, test design, model and pipeline failure analysis, pytest, GitHub Actions, CI/CD
- **Engineering:** FastAPI, REST APIs, Docker, Git, Linux

LANGUAGES

English: C1 | German: B1 completed; preparing for B2 exam | Arabic: Native